# PRAMANEX ClaimGraph-RD: A Temporal Patent Evidence Graph

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Engineering Provenance-Aware Trust in Rare-Disease and Biotech Intellectual Property Triage

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PRAMANEX / INTELLECTUAL PROPERTY SYSTEMS REFERENCE ARCHITECTURE / SYNTHETIC EVIDENCE

## EXECUTIVE VALUE ABSTRACT

Patent intelligence for rare-disease and drug-development programs is a reconstruction problem. A relevant limitation can sit in a continuation, a national phase, an amended claim or a jurisdiction-specific legal state. Scientific identity can appear as a gene, sequence, molecule, formulation, dose, indication or patient subgroup. A pure similarity score cannot resolve which legal right, at which time, might be material to a proposed product.

ClaimGraph-RD proposes a temporal, provenance-aware graph that binds candidate claim elements to family, jurisdiction, effective date, product features and exact source evidence. The reported study is a controlled synthetic benchmark rather than live patent coverage. Its author-reported ranking results motivate further development, while a counter-result shows that a generic hybrid can extract claim elements more accurately after retrieving less relevant material.

The industrial architecture therefore separates discovery, legal-state reconstruction, limitation mapping and qualified review. A future live ingestion network across patent offices and chemical/biological resources remains a roadmap requiring source rights, latency assessment and legal review. The system should accelerate evidence preparation and expose uncertainty; it cannot issue autonomous freedom-to-operate, validity or infringement conclusions. Its durable asset is a versioned graph of claims and evidence with reproducible updates and human dispositions.

| EVIDENCE | TEMPORAL | FAILURE-SAFE |
|---|---|---|
| Exact passages + versioned provenance | Jurisdiction + cutoff-state reconstruction | HOLD / quarantine for unresolved evidence |

**ENGINEERING THESIS** The graph and its provenance should outlast any one ranking model. Trust comes from making contradictions, missed scope, temporal state and human judgment inspectable.

Publication posture. Source-grounded reference architecture. The reported performance is synthetic and not independently reproduced; live legal-status accuracy, exhaustive search, counsel agreement and predictive FTO remain deployment questions.

## 1. THE GLOBAL PATENT RECONSTRUCTION PROBLEM

Patent documents contain families, priority claims, continuations, national phases and evolving claim text. Status can differ by jurisdiction and date. Scientific language adds a second mismatch: a sequence or chemical structure may be associated with a therapeutic indication under terminology absent from a product team's initial query.

The task is to find relevant candidates, locate limitations, reconstruct family and legal state at a declared cutoff, and preserve the evidence for counsel or a qualified analyst.

A query therefore specifies product features, scientific identities, jurisdictions, question type and time cutoff. The architecture is for high-sensitivity research triage and review preparation. It does not claim exhaustive search, legal clearance, non-infringement, validity or professional advice. Any live matter needs an agreed search scope and qualified professional judgment.

**STATE: INSUFFICIENT_SPECIFICATION**

Condition: one or more required query fields are absent or ambiguous.

Required boundary: product features / scientific identities / jurisdictions / question type / time cutoff.

Effect: scope remains unresolved; the system must not convert missing scope into a clearance statement.

## 2. GLOBAL PATENT GRAPH: TARGET ARCHITECTURE

The target graph distinguishes Publication, Application, Patent, Family, Claim Version, Limitation, Legal Event, Jurisdiction and Scientific Identity. Edges carry source reference, time semantics and extraction provenance. A claim text and its amendment are separate versioned objects; an event observed late does not silently rewrite the earlier state.

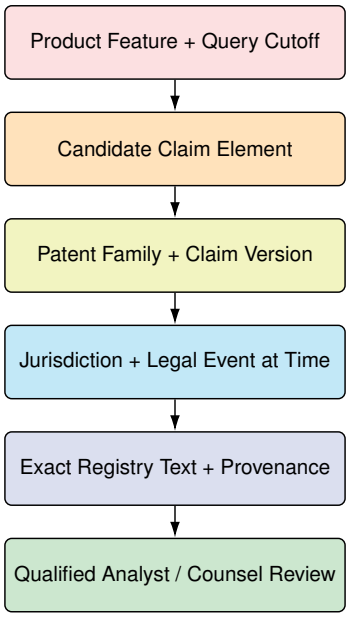
Identity links can connect genes, formulations, dosages, disease terms and chemical structures, but synonym or structure proximity is only a retrieval clue.

WIPO, USPTO and EPO records, plus authorized national-office feeds, are candidate source networks for a future live implementation.

Patent-office APIs and bulk data vary in scope, update latency, legal status and use rights. Chemically aware resources such as SureChEMBL and PatCID can enrich candidate discovery when their licenses and provenance are honored. No live, real-time, multi-jurisdictional integration is evidenced by the supplied synthetic study.

The ingestion layer must document connector coverage and freshness per source. Unavailable or ambiguous status remains unknown, not silently inferred.

**Figure 1. Claim Evidence Graph**




Proposed architecture | failures enter HOLD / quarantine

The graph preserves separate publication, application, family and legal-event nodes. A claim version does not inherit today's status when the question asks about an earlier cutoff. Source or status conflict remains visible, and the reviewer can inspect the passage behind each edge.

## 3. PROVENANCE-AWARE EXTRACTION FLOW

The pipeline begins with a matter definition and a controlled search plan. Candidate retrieval combines transparent lexical and representation signals with identity expansion; every candidate keeps its query, score basis and original source.

Family resolution groups related records without merging distinct legal rights. Temporal resolution reconstructs claim versions and observed legal events at the requested cutoff, retaining contradictory or missing states.

Claim-element mapping links a limitation to the product feature and exact patent passage. A verification gate checks that the source, date, jurisdiction, version and passage support the asserted relationship. Finally, a qualified reviewer accepts, corrects, requests evidence or rejects the candidate with rationale.

Generative language can assist bounded summarization but cannot supply absent legal facts. A prompt does not replace a deterministic source/version check.

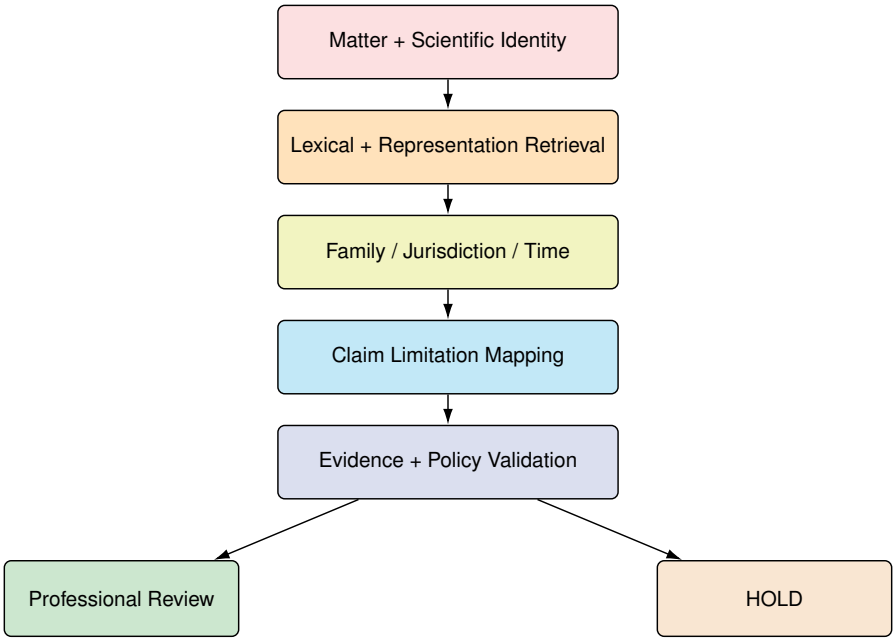
A stale citation, unsupported identity jump or unresolved family edge yields HOLD. A conflict between registry feeds remains visible with provenance and observation times. An update creates a new graph revision, invalidates affected downstream analyses and requests renewed human review. Prior reasoning remains reconstructable.

Structural invariant. Unsupported identity jumps or unresolved family configurations are first-class failure states. They cannot be normalized into a confident answer by narrative generation.

**HOLD / QUARANTINE TRIGGER**

Stale source • unsupported identity jump • unresolved family edge • missing cutoff-state evidence • source/status conflict without resolution

**Figure 2. Candidate-to-Review Failure-Safe Path**




Proposed architecture | failures enter HOLD / quarantine

A ranking model can nominate a record; it cannot infer legal materiality from textual similarity. An unsupported identity jump, ambiguous family or missing cutoff-state evidence cannot be rephrased as clearance.

## 4. ZERO IMPLICIT TRUST IN LEGAL INTELLIGENCE

Core system inequalities are treated as engineering constraints, not presentation slogans. Each one blocks a distinct class of legal-intelligence failure.

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TextSimilarity(claim, product) != LegalMateriality(claim, product)

Network Location (source) != AccessPermission(user, matter)

Citation != Applicability

CurrentStatus != HistoricalStatus
```

A high semantic match neither proves claim construction nor establishes scope, validity or jurisdictional effect. A ranking score can prioritize review; it cannot transform textual proximity into a legal conclusion.

A reachable record or internal search index does not authorize use of privileged work product. Matter boundaries, source rights and internal permissions must be evaluated separately from technical reachability.

A correctly linked paragraph can still refer to the wrong amendment or cutoff. Exact evidence is necessary, but applicability must be resolved against version, jurisdiction, date and matter question.

A live registry display cannot automatically answer the legal state at a prior date. Historical reconstruction is a time-indexed evidence problem.

Every matter requires source rights, tenant and privilege separation, least-privilege reviewer access, immutable or reconstructable version history, controlled export and audit of human decisions.

These principles align with WIPO patent-landscape methodology as an evidence discipline, but a system is not certified or legally sufficient because it cites WIPO. Access and search coverage must be tested under the intended organization's security and professional process.

**SECURITY & PRIVILEGE ARCHITECTURE**

Least-privilege review access | Immutable/reconstructable history | Tenant separation | Controlled export | Human-decision audit

## 5. EMPIRICAL EVIDENCE FROM THE SYNTHETIC STUDY

The source describes 30 generated repetitions, 252 synthetic patent records per repetition and 84 held-out queries. It reports family-level partitioning and a temporal holdout, but does not provide generator code, split manifests, seeds or per-run outputs. Numerical findings are author-reported and were not independently reproduced.

Four systems have reported outcome tables: lexical TF-IDF, character dense proxy, generic hybrid and ClaimGraph-RD. Other planned baselines and ablations without outcome tables are not treated as executed.

Reported Recall@5 is 0.224, 0.216, 0.375 and 0.799 respectively; nDCG@10 is 0.247, 0.243, 0.437 and 0.834; MRR is 0.339, 0.331, 0.438 and 0.840.

The full method's temporal-status accuracy of 1.000 reflects deterministic synthetic event rules, not live legal-status accuracy. Removing graph features reduces Recall@5 to 0.685; removing temporal reasoning reduces temporal-status accuracy to 0.752. The method's own Recall@5 remains below 0.80 in the synthetic setting, so missed relevant material is a central risk.

### 5.1 FULL REPORTED METRICS AND 95% CONFIDENCE INTERVALS

| Architecture | Recall@5 | nDCG@10 | MRR | Temporal-Status Accuracy |
|---|---|---|---|---|
| Lexical TF-IDF | 0.224 (0.210-0.238) | 0.247 (0.235-0.259) | 0.339 (0.323-0.356) | 0.744 (0.722-0.765) |
| Character Dense Proxy | 0.216 (0.202-0.230) | 0.243 (0.229-0.257) | 0.331 (0.314-0.348) | 0.721 (0.704-0.738) |
| Generic Hybrid | 0.375 (0.357-0.394) | 0.437 (0.419-0.456) | 0.438 (0.420-0.457) | 0.739 (0.721-0.756) |
| **ClaimGraph-RD** | **0.799** (0.775-0.823) | **0.834** (0.809-0.859) | **0.840** (0.814-0.865) | **1.000** (1.000-1.000) |

Reported Recall@5 differences: ClaimGraph-RD vs lexical TF-IDF = 0.575 (95% CI 0.552-0.598); vs generic hybrid = 0.424 (0.402-0.446). Reported nDCG@10 differences are 0.587 (0.565-0.609) and 0.396 (0.380-0.412), respectively. Per-run outputs and analysis code were unavailable, so the confidence-interval method and p-values could not be independently recalculated.

### 5.2 ABLATIONS, EXTRACTION QUALITY, AND THE CRITICAL COUNTER-RESULT

The reported ablations show that graph, identity and temporal components contribute differently to retrieval, ranking, mapping and legal-state reconstruction. These metrics must remain separated in evaluation.

| System / Ablation | Recall@5 | nDCG@10 | Claim-Element F1 | Temporal-Status Accuracy |
|---|---|---|---|---|
| Full ClaimGraph-RD | 0.799 | 0.834 | 0.963 | 1.000 |
| Without temporal reasoning | 0.760 | 0.782 | 0.973 | 0.752 |
| Without identity | 0.756 | 0.779 | 0.934 | Not reported |
| Without graph features | 0.685 | 0.724 | 0.895 | Not reported |
| Generic Hybrid | 0.375 | 0.437 | **0.999** | 0.739 |

**CRITICAL COUNTER-RESULT**

The generic hybrid's claim-element F1 is reported at 0.999, above ClaimGraph-RD's 0.963, despite substantially weaker retrieval and temporal results. Excellent extraction of the wrong family or cutoff state can be misleading. A single extraction metric cannot certify a legal-research workflow.

Without graph features, reported Recall@5 drops by 0.114 (95% CI 0.106-0.123). Without temporal reasoning, Recall@5/nDCG@10/claim-element F1 are 0.760/0.782/0.973 while temporal-status accuracy falls to 0.752. Without identity, the reported trio is 0.756/0.779/0.934.

Perfect synthetic temporal status comes from deterministic event rules and must not be generalized to live registry accuracy.

Evaluation must separately score discovery, family/temporal reconstruction, claim-limitation mapping, provenance and professional review. The architecture treats these as separable control surfaces rather than a single model score.

The synthetic benchmark is therefore a design signal, not production proof. Missed relevant material remains a central risk because the full method's reported Recall@5 is 0.799 in the controlled setting.

**EVALUATION RULE**

Never collapse retrieval quality, legal-state reconstruction, claim-element extraction and reviewer correction into one aggregate score. Each can fail independently.

## 6. REVIEW AND FAILURE-SAFE CONTRACT

The system must distinguish source unavailable, no candidate, ambiguous family, conflicting status, unsupported mapping and reviewer rejection. It cannot collapse these states into 'clear.' Material outputs carry exact passage, source/version, jurisdiction, date, query and limitations.

A reviewer sees competing evidence and the source freshness window, not just a narrative summary. Revision-bound approval expires when a dependent source or matter assumption changes.

Professional review is the decision gate, not an optional checkbox. An adverse case must be visible in the product's ordinary workflow.

**FAILURE-STATE CONTRACT**

UNKNOWN is not CLEAR. HOLD is not REJECTED. A source outage is not 'no result.' Conflicting status is preserved until adjudicated.

### 6.1 EXECUTED VERSUS PLANNED COMPARISONS

Four principal systems and three reported ablations have supplied result summaries. BM25, chemical/sequence-only retrieval, a human manual workflow, reranker removal, provenance-validator removal and specialist-decomposition removal appear in the broader protocol without corresponding outcome tables. They remain planned.

The synthetic failures include near-duplicate disease/modality combinations, mismatched product features and queries missing gene identity. Even in this controlled setting, average Recall@5 below 0.80 rules out exhaustive-search language.

**FUTURE LIVE PILOT - PRESPECIFIED VALIDATION PROTOCOL**

1. Prespecify search scope and cutoff. 2. Capture registry snapshots and source rights. 3. Freeze an independent answer key. 4. Measure missed material families, reviewer corrections, legal-state lag and reconstruction time. 5. Evaluate OCR, translation, Markush, sequence and continuation edge cases. 6. Obtain qualified counsel or analyst adjudication.

## 7. REPRODUCIBILITY AND DEPLOYMENT GATES

Independent replication of the reported synthetic benchmark needs generator code and parameters, generated fixtures or deterministic regeneration instructions, seeds, family and temporal split manifests, per-run outputs, analysis scripts, dependency versions, artifact hashes and documented reviewer annotations if reviewer claims are made. Until then, the benchmark is a design signal rather than production proof.

A controlled live build would first implement source/claim versioning and exact evidence links; then family, identity and temporal edges; then bounded ranking and review.

Deployment must prove permission denial, source-outage behavior, stale-analysis invalidation, correct deep links, audit replay and backup recovery.

Expansion into more jurisdictions depends on measured coverage and legal-state latency. No public repository should include a privileged client matter or unlicensed patent dataset.

## 8. STRATEGIC TECHNOLOGY HORIZON

### NEAR HORIZON

A reproducible public benchmark and narrow evidence-linked triage workflow with independently reviewed errors.

### MIDDLE HORIZON

Permissioned patent-office ingestion, chemical and biological identity indexing, temporal updates and human-correction analytics across declared jurisdictions.

### LONG HORIZON

A decentralized, continuously updated global FTO research network that predicts where new evidence merits qualified review. Prediction and autonomy describe research ambitions, not authority to render legal opinions.

### DEPLOYMENT GATE MATRIX

| Gate | Minimum proof before expansion |
|---|---|
| Evidence | Exact passage + source/version + date + jurisdiction preserved |
| Temporal | Historical cutoff reconstructed without inheriting current status |
| Permission | Denied sources and privileged matter boundaries enforced |
| Failure | Outage / ambiguity / conflict enters explicit UNKNOWN or HOLD |
| Audit | Revision history, human disposition and stale-analysis invalidation replayable |
| Coverage | Jurisdiction expansion tied to measured source coverage and status latency |

## 9. FIRST LIVE EVALUATION PROTOCOL

Select a limited set of jurisdictions and matter types with documented source rights. Freeze the cutoff date, search scope, product-feature descriptions and independent analyst answer key before system output is shown.

Measure relevant-family recall, temporal-state correctness, exact-passage grounding, unsupported mappings, reviewer correction burden and missed material documents. Report status latency for every source.

Test continuation, amended claim, translation, OCR, Markush, sequence and withdrawn/expired-right cases. Qualified counsel adjudicates legal significance. The study can evaluate triage usefulness but cannot certify an exhaustive FTO opinion.

## 10. STUDY INTEGRITY AND DECLARATIONS

The reported experiment used generated synthetic patent records, without human participants, patient data, privileged client matters or live legal analysis. Generative AI assisted language/structure editing; the author remains responsible for the scientific claims.

The source says deterministic Python generated outputs but supplies no code to check that statement. The author is developing the concept, a potential competing interest, and reports no external funding. A future live study needs source rights, confidentiality assessment and qualified professional authority.

No generator, seeds, split manifest, per-run outputs or reviewer annotation package accompanied the source. Thus a production extraction protocol, real-time WIPO/USPTO/EPO coverage, counsel agreement, predictive FTO network or legal clearance is a target to test rather than a demonstrated outcome.

### BOUNDARY OF CLAIM

ClaimGraph-RD is presented as a provenance-aware research and review architecture. It is not an autonomous legal-opinion engine, and the supplied evidence does not establish exhaustive live patent coverage or production legal-status accuracy.

## REFERENCES

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## DOCUMENT CONTROL & PUBLICATION NOTES

Evidence basis: controlled synthetic benchmark reported in the source architecture draft; not independently reproduced.

Human authority: qualified analyst/counsel review remains the decision gate. Core invariant: unresolved identity, family, temporal or source conditions remain visible as UNKNOWN/HOLD rather than being normalized into a legal conclusion.

PRAMANEX ClaimGraph-RD is an architecture for reconstructable evidence, not a substitute for professional legal judgment. The system thesis is that provenance, time and reviewability are control-plane properties: they determine whether a model output can be responsibly inspected, corrected and reused.